Fairness in Credit-Risk Classification

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GitHub repository: https://github.com/davisjoseph6/fairness-lab.git

Introduction and methodology

This report audits fairness in a binary credit-risk classifier trained on the German Credit dataset. The target is encoded as good credit = 1 and bad credit = 0. The sensitive attribute is age, divided into young applicants, defined as age below 25, and old applicants, defined as age 25 or above.

The experiment studies three families of fairness. Demographic parity belongs to independence and asks whether groups receive positive predictions at similar rates. Equalized odds belongs to separation and asks whether error rates are similar across groups conditional on the true label. Positive predictive value and calibration belong to sufficiency and ask whether a positive prediction means the same thing across groups.

The notebook uses one fixed train/test split, one random seed, one positive-class definition, and one age-group definition. Age is retained separately for auditing even when it is removed from the predictive features. This distinction is central to the lab: deleting a sensitive column from the model is not the same thing as proving that the model no longer produces group disparities.

Repository and submission artifacts

The full project repository and submission artifacts are available here:

- Repository: https://github.com/davisjoseph6/fairness-lab.git
- Pre-computation commitments: https://github.com/davisjoseph6/fairness-lab/blob/main/00_commitments.md
- Runnable notebook: https://github.com/davisjoseph6/fairness-lab/blob/main/notebooks/fairness_lab.ipynb
- Exported notebook HTML: https://github.com/davisjoseph6/fairness-lab/blob/main/outputs/fairness_lab.html
- Output tables: https://github.com/davisjoseph6/fairness-lab/tree/main/outputs/tables
- Output figures: https://github.com/davisjoseph6/fairness-lab/tree/main/outputs/figures
- Final report PDF: https://github.com/davisjoseph6/fairness-lab/blob/main/report/fairness_report.pdf

1. Baseline disparity

Our pre-computation expectation was that applicants below 25 would be disadvantaged and that deleting age would not eliminate the disparity. The experiment supported that expectation.

In the baseline model with age included, the old group had a higher base rate of good credit than the young group: 0.727 compared with 0.553. The model also selected old applicants more often: the old selection rate was 0.771, while the young selection rate was 0.660, giving a demographic-parity difference of 0.111.

The disadvantage was not limited to selection rate. The young group had a lower true-positive rate, 0.731 compared with 0.870 for the old group, meaning that creditworthy young applicants were less likely to be approved. The young group also had a higher false-positive rate, 0.571 compared with 0.507, and a lower positive predictive value, 0.613 compared with 0.821. Therefore, the answer to "who is disadvantaged" depends partly on the metric, but the young group is worse off on the main opportunity and prediction-quality measures in this run.

Removing age from the predictive features did not solve the problem. The demographic-parity difference remained exactly 0.111, and the equalized-odds difference slightly worsened from 0.139 to 0.144. This rejects the claim that a model cannot discriminate if it does not see the protected attribute. In this dataset, other variables such as employment, housing, credit duration, credit history, savings, and foreign-worker status may plausibly preserve age-related information as proxies. Removing age mostly removes the explicit variable; it does not remove all age-related structure from the data.

2. Calibration and equal-error impossibility

Using the baseline model, we verified Chouldechova's identity separately for the old and young groups. For the old group, the observed false positive rate was 0.507246 and the right-hand side of the identity was also 0.507246, with only floating-point error. For the young group, the observed false positive rate was 0.571429 and the identity also returned 0.571429.

The base rates were not equal. The old group had a positive-class base rate of 0.727273, while the young group had a base rate of 0.553191. This difference matters because the identity links the false positive rate to the base rate, the positive predictive value, and the false negative rate.

To demonstrate the conflict numerically, we used the pooled baseline PPV, 0.792035, as a common PPV value for both groups. Under that common PPV, while holding the observed false negative rates fixed, the identity forced the old group's FPR to 0.608858 and the young group's FPR to 0.237563. The resulting FPR gap was 0.371294.

Therefore, the conflict is not merely a bug in our classifier. It is a counting relationship. When groups have different base rates, equal calibration or equal PPV does not generally coexist with equal error rates. A stakeholder demand for both equal calibration and equal FPR is therefore mathematically incompatible in ordinary non-perfect settings.

3. Mitigation and fairness–accuracy frontiers

The mitigation results show that "make it fair" is not a complete technical instruction. Each method changed a different fairness family.

The baseline model without age had accuracy 0.746667, demographic-parity difference 0.111177, and equalized-odds difference 0.144231. Reweighing kept the same accuracy, 0.746667, while reducing demographic-parity difference to 0.076528 and equalized-odds difference to 0.111801. This was the best equalized-odds result among the main methods.

Among the main named mitigation methods, ExponentiatedGradient with a demographic-parity constraint produced the highest accuracy, 0.766667, and almost eliminated the demographic-parity difference, reducing it to 0.000673. In the epsilon sweep, the related EG-DP eps=0.2 point reached accuracy 0.770000 with demographic-parity difference 0.003280, but its equalized-odds difference remained 0.173913. This reinforces the same trade-off: the strongest demographic-parity points did not also minimise equalized odds.

For the main ExponentiatedGradient demographic-parity model, the improvement in selection parity came with a larger false-positive-rate gap: the old group's FPR was 0.492754, while the young group's FPR was 0.666667. This shows that improving independence did not automatically improve separation.

ExponentiatedGradient with an equalized-odds constraint did not perform best on equalized odds on the test set. It achieved accuracy 0.750000 and demographic-parity difference 0.003280, but its equalized-odds difference was 0.192547. The epsilon sweep also showed that the EG-EO points kept equalized-odds difference around 0.192547 on this test set. This reminds us that a constraint fitted on the training data does not guarantee the smallest test-set fairness gap.

ThresholdOptimizer with demographic parity achieved accuracy 0.740000, demographic-parity difference 0.074847, and equalized-odds difference 0.283644. ThresholdOptimizer with equalized odds achieved accuracy 0.733333, demographic-parity difference 0.005214, and equalized-odds difference 0.157350. Both threshold-optimisation methods use group-specific decision rules, so they must be interpreted differently from a single common classifier.

The two frontier plots also show that demographic parity and equalized odds are not one single fairness axis.

No method eliminated both fairness gaps without cost. The smallest demographic-parity difference among the main methods came from ExponentiatedGradient with DemographicParity, but its equalized-odds difference worsened relative to the baseline without age. The smallest equalized-odds difference among the main methods came from Reweighing, but it did not eliminate demographic-parity difference. The two frontier plots therefore represent different trade-offs rather than one universal fairness frontier.

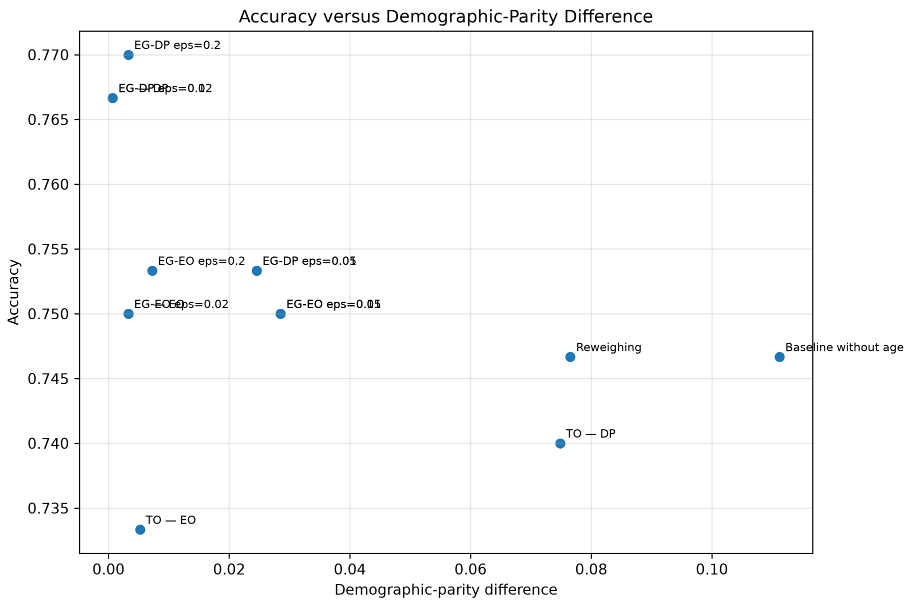

Figure 1: Accuracy versus demographic-parity difference

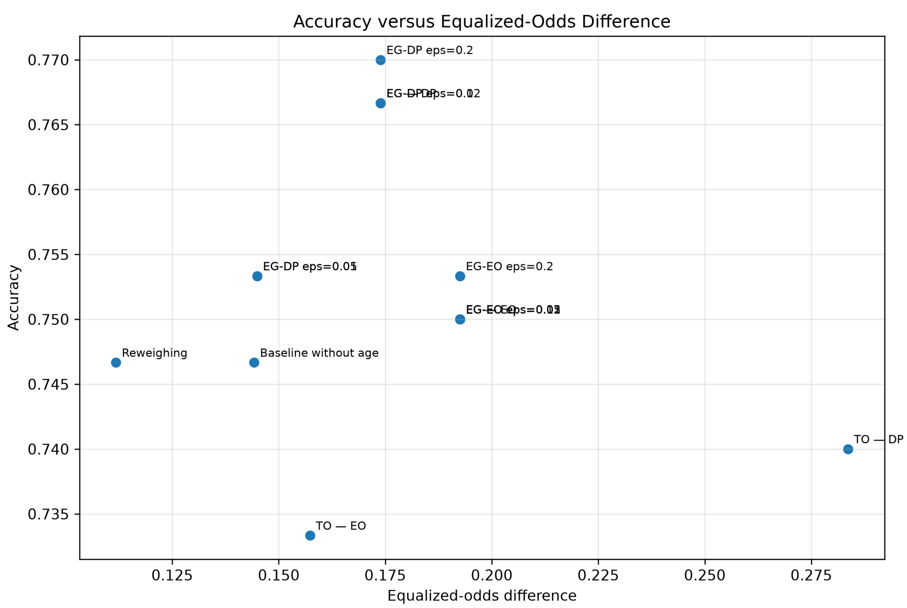

Figure 2: Accuracy versus equalized-odds difference

4. Chosen operating point

We choose the Reweighing operating point. We do not claim that it is the fairest model in general. We choose it because it best matches our normative priority: reducing unequal error rates and reducing the false-rejection burden on creditworthy young applicants, while avoiding a fully automated deployment decision.

Compared with the baseline without age, Reweighing kept accuracy unchanged at 0.746667. It reduced demographic-parity difference from 0.111177 to 0.076528 and reduced equalized-odds difference from 0.144231 to 0.111801, the smallest equalized-odds difference among the main methods. It also improved the young group's true-positive rate from 0.730769 to 0.769231, meaning that more creditworthy young applicants were approved.

The cost is not zero. The young group's false-positive rate increased from 0.571429 to 0.619048, and its positive predictive value decreased slightly from 0.612903 to 0.606061. In human terms, this means that the system may approve more young applicants who later turn out to be bad credit risks. We assign more of that residual cost to the bank through manual review, monitoring, and limited additional credit risk, rather than assigning it entirely to creditworthy young applicants who would otherwise be wrongly rejected.

We also reject a purely metric-based deployment decision. The label "good credit" is itself historical. It reflects past lending and repayment patterns, not a neutral moral truth. Applicants who were previously denied credit may be missing from the outcome data, and past credit access may have shaped the target itself. Therefore, this model should not be treated as a self-justifying decision-maker. At most, it could be used as a decision-support tool with human oversight, reason codes, and an appeal route.

5. Regulation and deployment limits

We would not deploy the chosen model as a fully automated credit decision system. If used at all, it should be used only as a decision-support tool with documentation, human review, bias monitoring, reason codes, and an appeal route.

Under the EU AI Act, an AI system used to evaluate the creditworthiness of natural persons or establish their credit score is a high-risk AI system, except where the system is used for detecting financial fraud. Our system is therefore high-risk if used for real creditworthiness assessment.

Two concrete obligations follow. First, the system requires data and bias governance. The provider would need to document the origin, preparation, assumptions, suitability, and representativeness of the training, validation, and test data. It would also need to examine possible biases and take appropriate measures to detect, prevent, and mitigate them. Our Mission 1 result shows why deleting age is not an adequate governance response: the demographic-parity difference remained 0.111177 after age was removed from the predictive features.

Second, the system requires human oversight. We would implement a manual review band and appeal route. All rejected applications, and all borderline cases within a predefined score interval, should be reviewed by a trained credit officer. The officer should receive the model score, reason codes, the applicant file, known model limitations, and a warning against automation bias. The reviewer must be able to override, reverse, or disregard the model output.

GDPR also limits deployment. A fully automated credit refusal may fall under the restriction on solely automated decisions that produce legal or similarly significant effects. Therefore, the system should not issue final automatic rejections. Rejected applicants should receive meaningful information, a reason code, the ability to express their point of view, human intervention, and a route to contest the decision.

Finally, age is not a GDPR Article 9 special-category attribute. Article 9 lists categories such as racial or ethnic origin, political opinions, religious or philosophical beliefs, trade-union membership, genetic data, biometric identification data, health data, sex life, and sexual orientation. However, age is still relevant for discrimination analysis. The legally and ethically defensible approach is not to delete age blindly, but to retain it in a controlled way for auditing and bias monitoring, with access controls and documentation. This reconciles the legal analysis with Mission 1: removing age from the model did not remove age-related disparity.

Conclusion

This lab shows why "make the model fair" is not a complete instruction. The baseline model disadvantaged young applicants under several metrics: young applicants had a lower base rate, lower selection rate, lower TPR, higher FPR, and lower PPV than old applicants. Removing age from the predictive features did not remove the disparity.

The impossibility result also appeared numerically. Chouldechova's identity reproduced the observed FPR in each group, and forcing a common PPV produced a large forced FPR gap because the groups had different base rates.

The mitigation results confirmed that fairness gains are transfers, not free lunches. ExponentiatedGradient with DemographicParity almost eliminated the demographic-parity gap, but worsened the equalized-odds gap. Reweighing gave the best equalized-odds result among the main methods without reducing accuracy, so we selected it as the most defensible operating point for our stated values.

Our deployment conclusion is cautious. The model should not be deployed as a fully automated credit-decision system. At most, it could be used as a documented decision-support tool with bias monitoring, human review, reason codes, and an appeal route.

References

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- Fairlearn documentation for MetricFrame, demographic parity, equalized odds, ExponentiatedGradient, and ThresholdOptimizer. https://fairlearn.org/